American Computer Science League

2023 Finals ● Short Problems Solutions ● Junior Division

1. Boolean Algebra

$$\overline{AB} + \overline{A(B + C)} = \overline{A} + \overline{B} + \overline{A} + \overline{B + C}$$
$$= \overline{A} + \overline{B} + \overline{B}\,\overline{C}$$
$$= \overline{A} + \overline{B}(1 + \overline{C})$$
$$= \overline{A} + \overline{B}$$

D. $\overline{A} + \overline{B}$

2. Boolean Algebra

$$A\overline{B} + A(\overline{B} + C) + \overline{B}(\overline{A} + \overline{C}) = A\overline{B} + A\overline{B} + AC + \overline{A}\,\overline{B} + \overline{B}\,\overline{C}$$
$$= A\overline{B} + \overline{A}\,\overline{B} + \overline{B}\,\overline{C} + AC$$
$$= \overline{B}(A + \overline{A} + \overline{C}) + AC$$
$$= \overline{B}(1 + \overline{C}) + AC$$
$$= \overline{B} + AC$$

To be FALSE, both terms must be 0. So $\overline{B} = 0$, $B = 1$.
$AC = 0$ means at least1 must be 0.
Therefore 010, 011 and 110 make it FALSE.

B. 3

3. Bit-String Flicking

010101 OR 101101 AND NOT 011011
= 010101 OR (101101 AND (NOT 011011))
= 010101 OR (101101 AND 100100)
= 010101 OR 100100
= 110101

C. 110101

4. Bit-String Flicking

(LCIRC-2 (RSHIFT-1 (NOT (LSHIFT-1 (RCIRC-2 11010)))))
= (LCIRC-2 (RSHIFT-1 (NOT (LSHIFT-1 10110))))
= (LCIRC-2 (RSHIFT-1 (NOT 01100)))
= (LCIRC-2 (RSHIFT-1 10011))
= (LCIRC-2 01001)
= 00101

A. 00101

5. Recursive Functions

$f(1) = 2$
$f(2) = 2*f(2-1) - 2 = 2*f(1) - 2 = 2*2 - 2 = 4 - 2 = 2$
$f(3) = 2*f(3-1) - 3 = 2*f(2) - 3 = 2*2 - 3 = 4 - 3 = 1$
$f(4) = 2*f(4-1) - 4 = 2*f(3) - 4 = 2*1 - 4 = -2 - 4 = -2$
$f(5) = 2*f(5-1) - 5 = 2*f(4) - 5 = 2*(-2) - 5 = -4 - 5 = -9$
$f(6) = 2*f(6-1) - 6 = 2*f(5) - 6 = 2*(-9) - 6 = -18 - 6 = -24$
$f(7) = 2*f(7-1) - 7 = 2*f(6) - 7 = 2*(-24) - 7 = -48 - 7 = -55$
$f(8) = 2*f(8-1) - 8 = 2*f(7) - 8 = 2*(-55) - 8 = -110 - 8 = -118$
So $f(f(f(f(8)))) = f(f(f(-118)))$
$= f(f(2))$ since $f(-118) = 2$
$= f(2)$
$= 2$

B. 2

6. Recursive Functions

$f(21) = 2 + f(21-3) = 2 + f(18) = 2 + 11 = 13$
$f(18) = 2 + f(18-3) = 2 + f(15) = 2 + 9 = 11$
$f(15) = 2 + f(15-3) = 2 + f(12) = 2 + 7 = 9$
$f(12) = 2 + f(12-3) = 2 + f(9) = 2 + 5 = 7$
$f(9) = f(9-2) - 1 = f(7) - 1 = 6 - 1 = 5$
$f(7) = f(7-2) = 1 = f(5) - 1 = 7 - 1 = 6$
$f(5) = 5 + 2 = 7$

D. 13

7. Digital Electronics

The circuit translates to: $\overline{(AB)(B + C)} + C$

$\overline{(AB)(B + C)} + C = \overline{(A1)(1 + C)} + C$ if $B = 1$
$= \overline{A1} + C$
$= \overline{A} + C$

B. $\overline{A} + C$

8. Digital Electronics

The Boolean expression represented by the following digital circuit is:

$\overline{A + \overline{AB} + BC} \oplus C = \overline{A + \overline{A} + \overline{B} + BC} \oplus C$
$= \overline{1 + \overline{B} + BC} \oplus C$
$= \overline{1} \oplus C$
$= 0 \oplus C$

This is TRUE when $C = 1$. Therefore **1 makes it TRUE.

C. 4

9. Prefix-Infix-Postfix

+ + 2 / * 3 − 4 1 3 ^ 2 3
= + + 2 / * 3 (− 4 1) 3 (^ 2 3)
= + + 2 / * 3 (4 1 −) 3 (2 3 ^)
= + + 2 / (* 3 (4 1 −)) 3 (2 3 ^)
= + + 2 / (3 4 1 − *) 3 (2 3 ^)
= + + 2 (/ (3 4 1 − *) 3) (2 3 ^)
= + + 2 (3 4 1 − * 3 /) (2 3 ^)
= + (+ 2 (3 4 1 − * 3 /)) (2 3 ^)
= + (2 3 4 1 − * 3 / +) (2 3 ^)
= (+ (2 3 4 1 − * 3 / +) (2 3 ^))
= 2 3 4 1 − * 3 / + 2 3 ^ +

B. 2 3 4 1 − * 3 / + 2 3 ^ +

10. Prefix-Infix-Postfix

3 2 # 5 4 # * 3 4 * − 4 2 # 1 − /
= (3 2 #) (5 4 #) * (3 4 *) − (4 2 #) 1 − /
= ((3−2)*(3+2))((5−4)*(5+4)) * (3*4) − ((4−2)*(4+2)) 1 − /
= (5 9 *) 12 − (12 1 −) /
= (45 12 −)11 /
= (33 11 /)
= 3

A. 3

11. Computer Number System

5272023_8 = 101 010 111 010 000 010 011_2
= 1 0101 0111 0100 0001 0011_2
= 1 5 7 4 1 3_{16}
= 157413

D. 157413

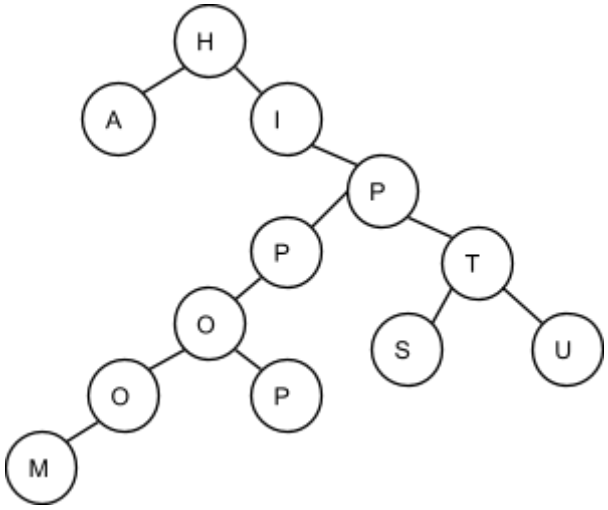
12. Computer Number Systems

$7_{10} + 77_8 + 777_{16}$
= $0111_2 + 111111_2 + 0111\ 0111\ 0111_2$
= $0111\ 1011\ 1101_2$
= BD_{16}

A. 7BD

13. Data Structures

The binary search tree for HIPPOPOTAMUS is:



The nodes at depth 3 are P and T.

C. PT

14. Data Structures

The queue is build as follows:

P, PR, R, RO, ROG, OG, OGR, OGRA, GRA, RA, RAM,
AM, AMM, MM, MME, ME, MER

The maximum number in the queue is 4.

D. 4

15. Graph Theory

If Miami is shut down, the following routes would be eliminated:
N → M, N → M → P, N → M → P → D, N → M → P → O,
N → M → P → C, N → M → P → C → O, N → P → M, and
N → C → P → M. That is a total of 8 routes.

D. 8

16. Graph Theory

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 1 & 2 & 1 \\ 2 & 1 & 1 & 2 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1 & 1 \end{bmatrix}$$

There are 8 pairs of vertices with no path of length 2 which are AA, AB, BE, CB, CD, CE, DA, and DE if done by inspection.

A. 8

17. What Does This Program Do?

30+40 > 50 so d = 16/2 = 8. 50-40 < 30 so d = 8%3 = 2. Then d = 2+1 = 3. 3%2 ≠ 0 and 3%3 = 0, so output (50+40) / 30 = 90 / 30 = 3.

A. 3

18. What Does This Program Do?

This program adds up the unique prime factors of 84. The main loop continues to divide possible factors, f, into the variable n until n < 2. When a factor is found in the inside loop, it is compared to the previous factor found and if not the same, it is added to the sum, s. Therefore, the unique factors are 2, 3, 7 which is a sum of 12.

D. 12

19. What Does This Program Do?

This program finds the number of swaps needed to put the list in sorted order. The two outside loops find the location of each number, 0 to 9, in the current array. The inside loop swaps the values in the locations between where the number should be and where it currently is and counts how many swaps are made. The following table shows the resulting array and the swaps made:

x	arr	cnt
original	8 6 2 9 3 5 7 1 4 0	0
0	0 4 1 7 5 3 9 2 6 8	5
1	0 1 4 7 5 3 9 2 6 8	6
2	0 1 2 9 3 5 7 4 6 8	9
3	0 1 2 3 9 5 7 4 6 8	10
4	0 1 2 3 4 7 5 9 6 8	12
5	0 1 2 3 4 5 7 9 6 8	13
6	0 1 2 3 4 5 6 9 7 8	14
7	0 1 2 3 4 5 6 7 9 8	15
8	0 1 2 3 4 5 6 7 8 9	16

B. 16

20. What Does This Program Do?

This program converts each 3-digit string to its octal equivalent from right to left. “111” = 7, “100” = 4, “010” = 2, “101” = 5 so the output is “5247”.

B. 5247